Table 1: Comparison on CLARE and CL-Drive under LOSO and 10-fold cross-validation. Values are mean ± std (%) over folds; $F1_M$ is macro-F1; Params is the trainable parameter count in thousands (K). Avg averages the two datasets within each protocol; rows ordered by mean macro-F1.

		CLARE						**CL-Drive**						**Avg**					
		LOSO			10-fold			LOSO			10-fold			LOSO			10-fold		
Model	Params	Acc	$F1_M$	AUC	Acc	$F1_M$	AUC	Acc	$F1_M$	AUC	Acc	$F1_M$	AUC	Acc	$F1_M$	AUC	Acc	$F1_M$	AUC
CNN	5679.5	73.0 ±16.0	47.4 ±5.7	53.1 ±13.1	74.3 ±9.2	46.9 ±4.2	57.5 ±6.5	64.3 ±12.4	54.6 ±9.3	60.3 ±12.4	62.6 ±9.3	52.3 ±8.1	57.0 ±10.2	68.6	51.0	56.7	68.4	49.6	57.2
Transformer	614.0	74.0 ±13.7	49.3 ±5.0	53.9 ±12.9	70.5 ±9.5	47.7 ±5.2	54.7 ±3.4	61.0 ±10.4	53.2 ±7.5	56.7 ±12.7	60.1 ±6.0	52.7 ±7.2	56.1 ±8.2	67.5	51.2	55.3	65.3	50.2	55.4
ResNet	1036.9	72.8 ±13.6	50.1 ±6.0	55.1 ±14.3	69.9 ±7.7	50.9 ±6.6	58.2 ±8.0	61.7 ±10.4	52.6 ±10.4	59.1 ±13.2	58.0 ±8.8	50.4 ±8.7	51.8 ±10.2	67.2	51.3	57.1	63.9	50.7	55.0
Uni-S4	221.2	76.7 ±12.2	56.4 ±11.3	61.8 ±17.8	77.0 ±7.0	52.5 ±6.7	59.6 ±6.5	70.4 ±10.3	49.3 ±7.9	56.9 ±10.6	69.9 ±7.0	46.0 ±5.5	54.8 ±7.7	73.5	52.8	59.3	73.4	49.2	57.2
Uni-Mamba	440.5	77.5 ±11.5	58.3 ±8.9	60.3 ±17.0	77.8 ±7.0	52.5 ±10.5	59.2 ±7.9	71.6 ±9.6	47.0 ±7.4	56.3 ±13.0	65.7 ±8.8	48.7 ±4.7	52.9 ±7.3	74.6	52.7	58.3	71.7	50.6	56.0
VGG	418.6	72.3 ±12.9	49.8 ±7.0	53.6 ±14.1	66.4 ±9.2	52.5 ±3.2	57.1 ±5.0	62.0 ±12.1	53.7 ±10.0	59.2 ±13.2	58.1 ±10.9	51.2 ±7.5	54.2 ±9.1	67.1	51.7	56.4	62.2	51.9	55.7
Uni-S4D	220.7	77.3 ±12.2	56.4 ±11.4	61.3 ±18.0	77.6 ±6.6	54.1 ±8.2	59.7 ±5.3	72.5 ±9.1	49.9 ±6.8	56.1 ±13.3	67.1 ±7.6	48.4 ±5.8	56.4 ±8.0	74.9	53.1	58.7	72.4	51.3	58.0
Bi-S4D	441.3	76.7 ±12.3	56.4 ±11.3	61.3 ±17.8	77.3 ±6.7	53.9 ±8.0	61.1 ±6.9	71.2 ±10.5	49.6 ±7.5	56.9 ±11.4	66.4 ±9.0	49.0 ±5.5	53.7 ±5.8	73.9	53.0	59.1	71.9	51.4	57.4
Bi-S4	442.4	77.2 ±12.0	56.7 ±11.4	61.9 ±18.2	77.7 ±6.7	53.0 ±9.2	60.2 ±7.8	71.7 ±9.7	51.9 ±8.8	58.0 ±12.9	66.1 ±9.0	49.5 ±5.2	54.1 ±5.5	74.4	54.3	60.0	71.9	51.2	57.2
Bi-Mamba	881.0	77.1 ±12.0	59.2 ±9.4	62.0 ±18.2	77.2 ±6.4	53.1 ±7.6	60.0 ±6.6	69.4 ±11.2	51.5 ±8.6	57.9 ±11.6	65.4 ±6.3	50.9 ±5.8	55.8 ±6.6	73.2	55.3	59.9	71.3	52.0	57.9

Table 2: On-device inference benchmark on NVIDIA Jetson platforms (batch size 1, sequence length 500, 30 input channels, 100 timed iterations after 20 warm-up iterations). Values are mean ± std over all trained fold checkpoints (60 per model per device). Size is the model size in MB and GFLOPs the analytic compute per inference (both checkpoint-independent); Lat. and P99 are mean and 99th-percentile per-inference latency; Perf is achieved GFLOP/s; RSS is host peak resident memory and GPU the peak GPU memory allocated (the large std for Mamba models reflects a one-time allocator warm-up peak in the first run of each session); Power is mean board power under load; E/Inf is energy per inference, Inf/J inferences per joule, and GFLOPS/W achieved compute per watt.

			Latency (ms)		**Throughput**		**Memory (MB)**			**Energy**		
Model	Size (MB)	GFLOPs	Lat.	P99	FPS	Perf (GFLOP/s)	RSS	GPU	Power (W)	E/Inf (mJ)	Inf/J	GFLOPS/W
Jetson Orin Nano												
CNN	21.67	0.154	4.44 ±0.08	7.92 ±0.65	225.3 ±4.1	34.7 ±0.6	797 ±12	38.1 ±0.0	5.65 ±0.06	25.1 ±0.5	39.9 ±0.8	6.15 ±0.13
Transformer	2.63	33.294	6.38 ±0.45	10.32 ±1.43	157.5 ±10.4	5245.0 ±346.5	719 ±6	13.7 ±0.0	6.41 ±0.16	40.8 ±2.0	24.6 ±1.1	817.63 ±37.12
ResNet	3.97	0.078	7.70 ±0.08	9.06 ±0.75	129.9 ±1.3	10.1 ±0.1	797 ±5	14.2 ±0.0	4.50 ±0.04	34.7 ±0.6	28.9 ±0.5	2.24 ±0.04
Uni-S4	0.84	0.009	20.26 ±0.13	24.22 ±0.45	49.4 ±0.3	0.4 ±0.0	853 ±14	44.2 ±0.0	13.37 ±0.09	271.0 ±0.7	3.7 ±0.0	0.03 ±0.00
Uni-Mamba	1.68	0.463	15.16 ±0.23	16.58 ±0.80	66.0 ±1.0	30.5 ±0.5	1410 ±4	31.3 ±64.3	4.95 ±0.66	74.9 ±8.8	13.5 ±1.1	6.23 ±0.52
VGG	1.60	0.078	3.95 ±0.05	5.11 ±0.51	253.2 ±3.3	19.7 ±0.3	742 ±5	12.3 ±0.0	4.20 ±0.06	16.6 ±0.4	60.4 ±1.3	4.69 ±0.10
Uni-S4D	0.84	0.009	10.89 ±0.12	13.01 ±1.17	91.8 ±1.0	0.8 ±0.0	806 ±3	41.9 ±0.0	8.54 ±0.07	93.1 ±1.5	10.7 ±0.2	0.10 ±0.00
Bi-S4D	1.68	0.018	21.16 ±0.20	22.59 ±0.73	47.3 ±0.4	0.8 ±0.0	836 ±5	43.0 ±0.0	9.80 ±0.04	207.4 ±1.9	4.8 ±0.0	0.09 ±0.00
Bi-S4	1.69	0.018	37.79 ±0.09	41.48 ±1.14	26.5 ±0.1	0.5 ±0.0	875 ±3	45.4 ±0.0	15.00 ±0.06	567.0 ±2.2	1.8 ±0.0	0.03 ±0.00
Bi-Mamba	3.36	0.925	30.31 ±0.42	31.62 ±1.14	33.0 ±0.5	30.5 ±0.4	1437 ±5	33.3 ±64.3	4.97 ±0.32	150.4 ±7.8	6.7 ±0.3	6.16 ±0.27
Jetson Orin NX												
CNN	21.67	0.154	5.99 ±0.19	7.80 ±1.28	167.2 ±5.3	25.8 ±0.8	761 ±13	38.1 ±0.0	6.98 ±0.07	41.8 ±1.1	23.9 ±0.6	3.69 ±0.10
Transformer	2.63	33.294	10.02 ±0.18	13.17 ±1.29	99.9 ±1.8	3325.3 ±60.9	683 ±9	13.7 ±0.0	7.09 ±0.05	71.0 ±1.0	14.1 ±0.2	468.74 ±6.63
ResNet	3.97	0.078	7.95 ±0.07	9.75 ±1.02	125.8 ±1.1	9.8 ±0.1	764 ±9	14.2 ±0.0	6.17 ±0.05	49.1 ±0.6	20.4 ±0.2	1.58 ±0.02
Uni-S4	0.84	0.009	48.76 ±0.64	53.80 ±2.14	20.5 ±0.3	0.2 ±0.0	826 ±12	44.2 ±0.0	9.72 ±0.38	473.7 ±12.4	2.1 ±0.1	0.02 ±0.00
Uni-Mamba	1.68	0.463	18.21 ±0.08	19.36 ±0.89	54.9 ±0.2	25.4 ±0.1	1135 ±12	31.3 ±64.4	6.32 ±0.32	115.1 ±5.7	8.7 ±0.4	4.03 ±0.18
VGG	1.60	0.078	4.19 ±0.03	4.99 ±0.50	238.8 ±1.7	18.5 ±0.1	702 ±7	12.3 ±0.0	5.80 ±0.06	24.3 ±0.3	41.2 ±0.5	3.20 ±0.04
Uni-S4D	0.84	0.009	11.87 ±0.12	15.06 ±1.77	84.2 ±0.9	0.8 ±0.0	799 ±15	41.9 ±0.0	9.53 ±0.23	113.1 ±3.4	8.8 ±0.3	0.08 ±0.00
Bi-S4D	1.68	0.018	22.54 ±0.13	24.59 ±2.11	44.4 ±0.3	0.8 ±0.0	830 ±15	43.0 ±0.0	10.87 ±0.18	245.0 ±5.0	4.1 ±0.1	0.07 ±0.00
Bi-S4	1.69	0.018	93.80 ±3.01	106.42 ±47.32	10.7 ±0.3	0.2 ±0.0	882 ±35	45.4 ±0.0	10.15 ±0.34	952.0 ±42.3	1.1 ±0.0	0.02 ±0.00
Bi-Mamba	3.36	0.925	36.52 ±0.40	38.20 ±1.35	27.4 ±0.3	25.3 ±0.3	1211 ±76	33.3 ±64.3	6.56 ±0.16	239.6 ±7.3	4.2 ±0.1	3.87 ±0.12
Jetson AGX Orin												
CNN	21.67	0.154	4.20 ±0.14	6.57 ±0.81	238.5 ±7.6	36.8 ±1.2	820 ±12	38.1 ±0.0	4.98 ±0.05	20.9 ±0.8	47.9 ±1.7	7.38 ±0.26
Transformer	2.63	33.294	7.18 ±0.81	10.00 ±1.05	140.9 ±15.2	4689.7 ±505.7	741 ±3	13.7 ±0.0	4.51 ±0.04	32.4 ±3.5	31.2 ±3.3	1038.57 ±109.93
ResNet	3.97	0.078	6.45 ±0.07	8.47 ±0.86	154.9 ±1.6	12.0 ±0.1	823 ±4	14.2 ±0.0	4.33 ±0.03	28.0 ±0.4	35.8 ±0.5	2.78 ±0.04
Uni-S4	0.84	0.009	26.05 ±0.08	30.33 ±1.62	38.4 ±0.1	0.3 ±0.0	890 ±2	45.0 ±0.0	7.37 ±0.01	191.9 ±0.6	5.2 ±0.0	0.05 ±0.00
Uni-Mamba	1.68	0.463	14.10 ±0.29	17.21 ±2.05	71.0 ±1.4	32.8 ±0.7	1288 ±1	31.3 ±64.3	4.51 ±0.23	63.6 ±3.7	15.8 ±0.8	7.29 ±0.39
VGG	1.60	0.078	3.30 ±0.04	4.53 ±0.49	302.8 ±4.1	23.5 ±0.3	768 ±3	12.3 ±0.0	4.27 ±0.03	14.1 ±0.2	70.9 ±1.1	5.50 ±0.09
Uni-S4D	0.84	0.009	9.52 ±0.10	13.31 ±0.97	105.1 ±1.1	0.9 ±0.0	850 ±2	42.6 ±0.0	6.26 ±0.06	59.5 ±0.8	16.8 ±0.2	0.15 ±0.00
Bi-S4D	1.68	0.018	18.37 ±0.39	20.12 ±1.25	54.5 ±1.0	1.0 ±0.0	885 ±7	43.8 ±0.0	6.70 ±0.01	123.1 ±2.5	8.1 ±0.1	0.15 ±0.00
Bi-S4	1.69	0.018	50.54 ±0.16	55.32 ±1.69	19.8 ±0.1	0.4 ±0.0	937 ±23	46.1 ±0.0	7.60 ±0.02	384.1 ±0.8	2.6 ±0.0	0.05 ±0.00
Bi-Mamba	3.36	0.925	28.22 ±0.69	29.92 ±1.56	35.5 ±0.8	32.8 ±0.8	1318 ±3	33.3 ±64.3	4.75 ±0.27	134.0 ±10.5	7.5 ±0.6	6.94 ±0.52